Individual Project · AI-Assisted

Empirical Paper Replication and Extension

1. Research question and motivation of the selected paper

The minimum-wage employment effect is a long-standing controversial topic in labour economics. The conventional competitive-market theory predicts that binding minimum wage hikes will reduce low-wage sector employment. Card & Krueger (1994) deliver a landmark natural-experiment study which challenges this traditional prediction by investigating New Jersey's 1992 minimum-wage increase from $4.25 to $5.05 per hour, using fast-food restaurants in eastern Pennsylvania as the control group.

This replication project is motivated by three considerations. First, minimum-wage policy remains highly relevant for modern labor-market debates, and this paper is a canonical teaching case for difference-in-differences methodology, which closely matches my applied-economics major training. Second, famous empirical papers are subject to replication scrutiny; reproducing its results helps me understand the sensitivity of causal estimates to sample filtering and variable construction. Third, beyond pure replication, I implement an independent heterogeneity extension to explore whether minimum-wage impacts differ by store ownership, to gain deeper insight into heterogeneous firm-level responses.

The core research question of the original paper: Does a large minimum-wage rise reduce employment among low-wage fast-food establishments?

2. Data sources, sample construction, and variable definitions

The original dataset comes from two-wave telephone survey data collected by Card & Krueger (1994). Wave 1 (Feb-Mar 1992, pre-policy) and Wave 2 (Nov-Dec 1992, post-policy) survey fast-food stores belonging to four chains: Burger King, KFC, Roy Rogers and Wendy's. The full initial sample frame contains 473 stores, and 410 valid responses are obtained in wave 1. Wave 2 collects follow-up information including store closure status, achieving near-complete follow-up for wave-1 respondents.

Sample construction follows the original paper's rules. Permanently closed stores assign wave-2 employment to zero; temporarily closed stores mark wave-2 employment as missing. For Table 4 regression analysis, we further restrict the sample to stores with non-missing employment and starting-wage data in both waves, yielding 351 observations in this replication versus 357 in the original paper.

Key variable definitions:

FTE (full-time-equivalent employment): full-time workers including managers + 0.5 × number of part-time workers.

ΔFTE: change of FTE employment, calculated as FTE_wave2 − FTE_wave1, the main dependent variable for regressions.

Treat (NJ dummy): equals 1 if store locates in New Jersey (treatment group), 0 for Pennsylvania (control group).

GAP: proportional required wage increase to reach the new minimum wage of $5.05. GAP = 0 for PA stores and NJ stores whose wave-1 starting wage ≥ 5.05; otherwise GAP = (5.05 − W_{1i}) / W_{1i}.

Company-owned dummy: equals 1 for company-operated stores, 0 for franchise-owned stores, used in control variables and heterogeneity extension.

3. Empirical design

This project applies difference-in-differences (DID), the core empirical framework of the original paper. The identification strategy relies on the quasi-natural experiment: New Jersey raised its minimum wage while Pennsylvania kept its minimum wage unchanged. Under the parallel-trend assumption, the employment change difference between the two states isolates the causal effect of the minimum-wage reform.

First, descriptive DID comparisons are implemented in Table 3. We compare the mean change of FTE across treatment and control groups, and also conduct within-New-Jersey comparison between high-wage and low-wage stores as a specification check.

Second, reduced-form OLS regression models are estimated, taking ΔFTE_i as the dependent variable. Two main model specifications are adopted:

$$\Delta FTE_i = \beta_0 + \beta_1 NJ_i + X_i'\gamma + \varepsilon_i$$

$$\Delta FTE_i = \alpha_0 + \alpha_1 GAP_i + X_i'\lambda + \mu_i$$

Where X_i contains chain dummies, company-owned dummy and regional dummies in corresponding columns. NJ_i is the state treatment dummy, and GAP_i captures treatment intensity by pre-reform wage level.

For the independent extension, I run split-sample regressions for company-owned and franchise-owned subsamples, alongside a pooled regression with treat × company-owned interaction term, to test whether employment responses vary systematically by ownership type. All specifications keep consistent sample cleaning rules with Table 4, forming a natural connection with the subsequent replication comparison and extension results.

4. Core replication results and comparison with the original paper

（1）Table 2 — Means of Key Variables

Table 2 replicates the pre-policy summary statistics of store-level variables for New Jersey and Pennsylvania. There are minor discrepancies in sample size and variable means between the replicated results and the original paper. The main source is subtle differences in sample filtering rules: the original paper's handling of missing values for temporarily and permanently closed stores, together with outlier trimming criteria, are not fully identical to our cleaning procedure. Nevertheless, the pre-treatment feature patterns of the two samples are consistent with the original work. The gaps between New Jersey and Pennsylvania stores are small, supporting the parallel-trend assumption. These deviations lie within an acceptable range.

（2）Table 3 — Mean FTE Employment by State and Wage Group

Table 3 reports mean FTE employment by state and by wage group, together with the average change between waves. Rows 1, 2, 4 and 5 replicate cell for cell at the precision the paper prints: mean FTE before the increase is 20.44 in New Jersey and 23.33 in Pennsylvania, after the increase 21.03 and 21.17; the change in mean FTE is +0.47 for New Jersey against −2.28 for Pennsylvania, giving a difference-in-differences estimate of +2.75 against the published +2.76. Row 3, the change in mean employment, also matches in its means (New Jersey +0.59, Pennsylvania −2.16), but our standard errors were initially 10 – 25% smaller than CK's (Pennsylvania 1.23 vs 1.25; New Jersey 0.47 vs 0.54; the wage-group columns 0.82 / 0.68 / 0.99 vs 0.95 / 0.84 / 1.14). This is not a sample-size problem but a variance-definition problem. Row 3 differences two means computed on the same, overlapping stores, so the variance of that difference is $se_1^2 + se_2^2 - 2\rho \cdot se_1 \cdot se_2$ rather than the sum of the two variances. The public-use extract's own within-store correlation between wave-1 and wave-2 FTE is 0.57 in New Jersey and 0.45 in Pennsylvania, but CK evidently imposed a single common correlation across all columns instead of using each state's own. Imposing one pooled $\rho = 0.45$ reproduces every published row-3 standard error to within 0.01 (fitted values 0.451, 0.450, 0.457, 0.452, 0.452), which confirms that the gap lay in the covariance convention and not in the underlying data.

（3）Table 4 — Reduced-Form Regressions of the Change in Employment

Table 4 reports reduced-form regressions of the change in FTE employment on the New Jersey dummy (columns i–ii) and on GAP (columns iii–v). Our estimates track the published ones closely and, more importantly, are statistically indistinguishable from them. The New Jersey dummy is 2.28 (1.19) against 2.33 (1.19) in column (i) and 2.28 (1.20) against 2.30 (1.20) in column (ii); the GAP coefficients are 17.05 (6.09), 16.36 (6.24) and 13.88 (7.44) against published 15.65 (6.08), 14.92 (6.21) and 11.91 (7.39). Every standard error replicates to within 0.05, and every point estimate lies within 0.27 of a standard error of its published counterpart, so no coefficient changes its magnitude class, and every coefficient that is significant at the 5% level in the published table remains significant at the 5% level in ours; the only borderline case is GAP in column (v), where the published t of 1.61 lies just below and ours of 1.87 just above the 10% critical value. The remaining differences sit in the sample rather than in the specification. Our estimation sample contains 351 stores rather than 357, so the restriction "employment and starting wages available in both waves" binds slightly harder in the public-use extract than it did on CK's own extracts; most plausibly the six extra stores CK kept are records for which wave-2 wage information is incomplete in the public file. Consistently with a handful of extreme observations being affected rather than a general change in the data, the standard deviation of the dependent variable matches to within 1% (8.75 against 8.83) and every standard error matches to within 0.05, while its mean shifts from −0.237 to −0.030 — a difference of less than 0.03 standard deviations of the dependent variable itself. The residual variance is equally close (standard error of the regression 8.71 / 8.72 / 8.66 / 8.68 / 8.67 against 8.79 / 8.78 / 8.76 / 8.76 / 8.75), and the joint F-tests on the control variables remain insignificant in every model ($p = 0.48$ / 0.62 / 0.48 against 0.34 / 0.44 / 0.40), preserving the paper's substantive conclusion that neither store characteristics nor regional controls add explanatory power.

（4）Figure 1 — Distribution of Starting Wage Rates

Figure 1 plots the distribution of starting wage rates in February 1992 and November 1992, with New Jersey in solid black and Pennsylvania hatched. Our reproduction captures both of the paper's qualitative signatures. In February 1992 the modal wage in both states is the $4.25 federal minimum — 32.5% of New Jersey stores and 34.2% of Pennsylvania stores, against approximately 32% and 34% in the published chart — with the remaining mass concentrated at $4.45–$4.55 and at $4.75, and the two states' distributions lying almost on top of one another, as they must before the policy change. In November 1992 the New Jersey distribution collapses onto a single point at $5.05, where 89.6% of stores pay the new state minimum (published: approximately 90%), while Pennsylvania's distribution is essentially unchanged, still spread across $4.25, $4.45, $4.75 and $4.95. That asymmetry is the clearest single piece of evidence for the treatment effect in the paper, and it is reproduced exactly. What remains are small bar-level differences of one to three percentage points, concentrated at the $4.45/$4.55 and $4.95/$5.05 boundaries — precisely where a bin-edge convention difference, that is, whether a printed label denotes the lower or the upper edge of a $0.10-wide bin, will show up — together with slightly different denominators (314 New Jersey and 76 Pennsylvania wage observations in wave 1; 318 and 71 in wave 2, against the paper's somewhat different counts). Neither changes the modal locations, the shape of either distribution, or the February-to-November comparison that the figure exists to display.

5. Independent Extension: Does Ownership Moderate the Employment Effect?

Research Objective

The difference-in-differences estimate in Table 3 and the reduced-form GAP regressions in Table 4 both report a single average treatment effect for New Jersey. An average can conceal systematic heterogeneity across stores. This extension asks whether the employment response to the minimum-wage increase differed between company-owned and franchisee-owned stores. The distinction is economically meaningful: a company-owned outlet belongs to a corporate chain whose wage policy is set centrally and whose operating standards constrain how the store may adjust staffing, whereas a franchisee is a residual claimant with direct control over hiring and, typically, thinner margins and less capacity to absorb a legislated cost increase. Theory does not deliver an unambiguous prediction. The mandate binds both types equally in legal terms, so the question of which type had more room, and more incentive, to adjust employment is an empirical one.

Empirical Specification

We use the same estimation sample and variable definitions as the Table 4 replication: 351 stores with valid wave-1 and wave-2 employment and starting-wage data, of which 285 are in New Jersey and 66 in Pennsylvania (100 New Jersey and 24 Pennsylvania stores are company-owned; 185 and 42 respectively are franchisee-owned). The dependent variable is the change in full-time-equivalent employment between waves, $\Delta FTE = FTE(1992{:}11) - FTE(1992{:}02)$, where FTE = managers + full-time workers + 0.5 × part-time workers, with managers counted as full-time employees and permanently closed stores assigned a wave-2 FTE of zero. Treatment is a New Jersey dummy. Ownership is a dummy equal to one for

company-owned stores (CO_OWNED = 1) and zero for franchisees. The control block contains chain dummies (KFC, Roy Rogers and Wendy's, with Burger King as the omitted category) and region dummies (SOUTHJ, CENTRALJ, PA1, PA2). Ownership is deliberately excluded from the controls, since it is the moderating variable of interest and would otherwise be collinear with the interaction term. Task A estimates $\Delta FTE = \alpha + \beta \cdot treat + controls$ separately on the company-owned (124 stores) and franchisee (227 stores) subsamples, letting every coefficient differ by ownership type. Task B pools the sample and estimates $\Delta FTE = \alpha + \beta_1 \cdot treat + \beta_2 \cdot own + \beta_3 \cdot (treat \times own) + controls$, which constrains the control coefficients to be common and attributes the entire heterogeneous response to β_3.

Independent extension: DID heterogeneity by store ownership
Dependent variable: change in FTE employment (wave 2 - wave 1)
Estimation sample : 351 stores (NJ 285, PA 66) -- Table 4 sample rules
Mean / SD of dep. var.: -0.030 / 8.748

NJ x company-owned n = 100 mean delta_fte = 0.180
NJ x franchisee n = 185 mean delta_fte = 0.516
PA x company-owned n = 24 mean delta_fte = -1.917
PA x franchisee n = 42 mean delta_fte = -1.857
raw (unconditional) DID : company-owned = 2.097 franchisee = 2.373

delta_fte	(A1) Company-owned	(A2) Franchisee-owned	(B) Pooled + interaction
treat (NJ dummy)	3.395***	1.026	1.541*
	(1.159)	(1.054)	(0.926)
own (company-owned)			-0.482
			(2.440)
treat x own			0.662
			(2.633)
chain: KFC	-0.885	0.007	0.326
	(1.958)	(1.784)	(1.320)
chain: Roy Rogers	-4.623**	-0.897	-2.164
	(1.780)	(2.043)	(1.336)
chain: Wendy's	-3.814	0.113	-0.316
	(3.026)	(1.854)	(1.541)
region: SOUTHJ	-2.753*	2.059	0.195
	(1.503)	(1.741)	(1.216)
region: CENTRALJ	-4.323*	-0.426	-1.299
	(2.301)	(1.794)	(1.412)
region: PA1	-4.707*	-2.231	-3.175**
	(2.491)	(1.660)	(1.332)
region: PA2	2.231	0.398	1.125
	(1.537)	(1.723)	(1.374)
N (observations)	124	227	351
R-squared	0.1503	0.0221	0.0335
SER	6.845	9.539	8.713

Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01
Constant estimated but reported in the detail panel below.

(A1) Company-owned N = 124 df = 116 R2 = 0.1503 SER = 6.845

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================================================================================
        Variable                 Coef.   Std.Err.    t-stat    p-value
--------------------------------------------------------------------------------
        const                    0.918      1.532     0.599     0.5501
      treat                    3.395      1.159     2.929     0.0041  ***
        chain: KFC              -0.885      1.958    -0.452     0.6520
      chain: Roy Rogers        -4.623      1.780    -2.598     0.0106  **
        chain: Wendy's          -3.814      3.026    -1.261     0.2100
       region: SOUTHJ          -2.753      1.503    -1.832     0.0694  *
       region: CENTRALJ        -4.323      2.301    -1.879     0.0628  *
       region: PA1             -4.707      2.491    -1.890     0.0613  *
        region: PA2              2.231      1.537     1.451     0.1494

  (A2) Franchisee-owned    N = 227    df = 219    R2 = 0.0221    SER = 9.539
================================================================================
        Variable                 Coef.   Std.Err.    t-stat    p-value
--------------------------------------------------------------------------------
        const                   -0.807      0.839    -0.961     0.3375
        treat                    1.026      1.054     0.974     0.3313
        chain: KFC               0.007      1.784     0.004     0.9969
        chain: Roy Rogers       -0.897      2.043    -0.439     0.6610
        chain: Wendy's           0.113      1.854     0.061     0.9515
        region: SOUTHJ           2.059      1.741     1.183     0.2382
        region: CENTRALJ        -0.426      1.794    -0.238     0.8124
        region: PA1             -2.231      1.660    -1.344     0.1803
        region: PA2              0.398      1.723     0.231     0.8173

 (B) Pooled + interaction    N = 351    df = 341    R2 = 0.0335    SER = 8.713
================================================================================
        Variable                 Coef.   Std.Err.    t-stat    p-value
--------------------------------------------------------------------------------
        const                   -0.509      0.751    -0.678     0.4981
       treat                    1.541      0.926     1.664     0.0971  *
        own                     -0.482      2.440    -0.197     0.8437
        treat x own              0.662      2.633     0.252     0.8016
        chain: KFC               0.326      1.320     0.247     0.8053
        chain: Roy Rogers       -2.164      1.336    -1.620     0.1062
        chain: Wendy's          -0.316      1.541    -0.205     0.8375
        region: SOUTHJ           0.195      1.216     0.160     0.8730
        region: CENTRALJ        -1.299      1.412    -0.920     0.3581
      region: PA1               -3.175      1.332    -2.384     0.0177  **
        region: PA2              1.125      1.374     0.818     0.4137
               READING THE INTERACTION TERM  beta3  (column B)
================================================================================
        beta3 (treat x own) = 0.662  (SE 2.633),  t = 0.252,  p = 0.8016
     implied DID effect, franchisee    (own = 0) : beta1        =   1.541
     implied DID effect, company-owned (own = 1) : beta1 + beta3 =   2.203
    beta3 is the ADDITIONAL employment effect of the New Jersey minimum-wage
     increase for company-owned stores relative to franchisee-owned stores.
   beta3 = 0 -> ownership does not moderate the policy effect (the two split-
   sample DID estimates would coincide); beta3 > 0 -> the effect is larger in
               company-owned stores; beta3 < 0 -> it is smaller.
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Results

The unconditional cell means show almost no heterogeneity. Average ΔFTE is +0.180 for company-owned New Jersey stores and +0.516 for franchisee New Jersey stores, against

−1.917 and −1.857 for their Pennsylvania counterparts, giving raw DID estimates of 2.10 for company-owned stores and 2.37 for franchisees — a gap of only 0.28 jobs, and in the direction opposite to the "company-owned adjusts more" story. Adding controls reverses the pattern. In the split-sample regressions the treatment effect is 3.395 (1.159) for company-owned stores, significant at the 1% level, and 1.026 (1.054) for franchisees, not significant; the point estimate is more than three times as large for company-owned stores. The pooled model, however, does not confirm this. The interaction coefficient is 0.662 with a standard error of 2.633, the franchisee effect — the coefficient on treat, now read as the reference category — is 1.541 (0.926), significant at the 10% level, and the implied company-owned effect is their sum, 2.203. The ownership dummy itself is small and insignificant (−0.482, SE 2.440). The formal test of heterogeneity is therefore unambiguous: β_3 is far from significance, and the difference between the two split-sample treatment effects, 2.37, carries an implied standard error of about 1.57 ($t \approx 1.5$), which is likewise insignificant at conventional levels. Crucially, the gap between the two specifications is traceable to the control coefficients rather than to the treatment effect. Roy Rogers carries a coefficient of −4.623 (significant at the 5% level) in the company-owned regression but only −0.897 in the franchisee regression, while KFC is insignificant in both (−0.885 with SE 1.958, and 0.007 with SE 1.784). The two subsamples therefore do not draw on the same chain mix, and the split-sample "ownership effect" partly reflects that composition gap. Every estimate of the DID effect in this extension is nonetheless positive — consistent with the paper's finding that the minimum-wage increase did not reduce employment — but the evidence that ownership moderates that effect is weak and specification-dependent.

Economic Interpretation and Limitations

Two readings are consistent with these results, and both should be stated. Taken at face value, the split-sample estimates suggest the policy effect was concentrated in company-owned stores, which would imply that centrally managed chains — less able to absorb the higher wage floor through the operating standards imposed by head office — adjusted employment more, while franchisees, as residual claimants with stronger incentives to economise on labour, had already been operating at leaner staffing levels and had less room left to cut. The pooled interaction model undercuts that story: once the control coefficients are constrained to be equal across ownership types, the estimated differential falls from 2.37 to 0.66 and is far from significance. The gap between the two specifications is itself the most informative result, and it arises because the split samples are not comparable. The company-owned subsample contains only 24 Pennsylvania stores, and its chain composition is unbalanced: the Roy Rogers dummy alone carries a coefficient of −4.623 in the company-owned regression against −0.897 in the franchisee regression, so a large part of the apparent "ownership" gap is in fact a chain-composition gap. With roughly 124 company-owned observations spread across four chains and two states, the split-sample contrast absorbs compositional differences that the pooled model does not credit. The conservative conclusion is therefore the correct one: point estimates remain positive throughout, but we cannot reject the null hypothesis that the minimum-wage increase affected company-owned and franchisee-owned stores equally. Several further limitations should be recorded. First, the design is thin where it matters: with 24 Pennsylvania company-owned stores supplying the counterfactual for 100 New Jersey company-owned

stores, the power to detect an interaction is low, and the confidence interval on β_3 spans essentially the whole range of economically plausible values. Second, our sample is the 351 stores available in the public-use extract rather than the 357 of the original paper, so the six-store discrepancy documented for Table 4 carries over into this extension. Third, ownership is measured at wave 1 only, and we cannot observe any change in ownership status between the two waves. Fourth, the design captures a single short post-treatment window and a single binary treatment indicator; the GAP intensity measure used in Table 4 is not exploited here, so the analysis cannot distinguish between "treated at all" and "treated more intensively". Given these constraints, the extension is best read as showing no evidence of systematic ownership heterogeneity, rather than as evidence that such heterogeneity is absent.

6. Validation, limitations, and conclusion

The core qualitative conclusions of this replication are consistent with Card & Krueger (1994). Replicated Table 2-4 and Figure 1 reproduce the original paper's main findings: the minimum-wage increase in New Jersey did not reduce fast-food FTE employment. Minor numerical discrepancies come from subtle missing-value filtering, causing a small sample-size gap, but do not overturn economic inference. The wage distribution in Figure 1 validates the first-stage effect: the policy was effectively enforced in New Jersey, with most stores raising starting wages to $5.05 after the reform.

Several limitations should be noted. First, this study only observes surviving or sampled fast-food stores; it cannot identify the minimum-wage effect on newly opened restaurants, which the original paper also acknowledges. Second, survey data inevitably contains measurement error in employment and wage variables, which may bring attenuation bias especially for the GAP-based specifications. Third, the heterogeneity extension shows larger positive point estimates for company-owned stores, yet the interaction term is statistically insignificant. Limited sample size, especially few Pennsylvania company-owned stores, lowers statistical power to detect group differences. Fourth, the analysis is restricted to the fast-food industry; conclusions cannot be mechanically generalized to all low-wage industries.

This replication reproduces the classic quasi-natural-experiment study by Card & Krueger (1994). Consistent with the original work, we find no evidence that minimum-wage growth lowers store-level fast-food employment. Though numerical deviations exist within acceptable replication tolerance, key qualitative results hold. The supplementary heterogeneity exercise suggests tentative evidence that company-owned stores may absorb labour-cost shocks better, but we cannot draw definitive statistical conclusions. This project demonstrates both the value and fragility of causal estimates derived from survey-based natural experiments: results are sensitive to sample-cleaning rules, and robustness checks are indispensable for empirical labour-market research.